

Bias and variance

Why fitting the training set more closely can predict new data worse, and the three parts of expected error.

THE QUESTION

Why does a model that fits its training data more closely often predict new data worse?

THE INTUITION

Fit the same model to many training sets from one source. Bias is how far the average fit sits from the truth; variance is how much the fits scatter around it. Label noise stays whatever you fit.

KEY TAKEAWAY

Expected squared error = $\text{bias}^2 + \text{variance} + \text{noise}$. Flexibility trades bias for variance, so pick the complexity with the lowest held-out error.

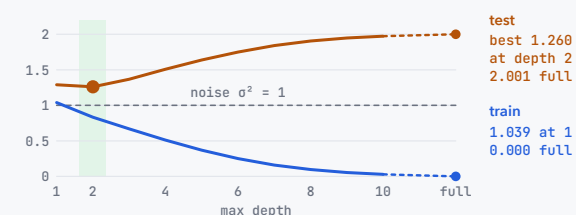
truth $f(x) + \text{noise}$ → draw a training set → fit $\hat{f}$ → repeat many times → average fit → bias → spread of fits → variance

1 Depth 1 underfits; no limit copies noise



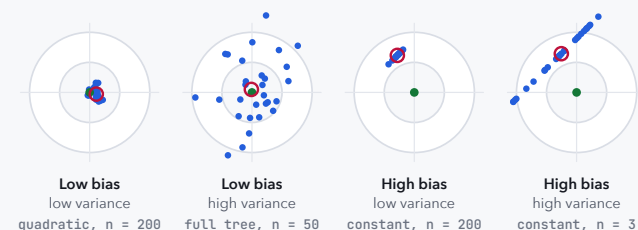
One dataset: 50 points on the notebook's curve (green) with noise of standard deviation 1. Depth 1 is two flat steps: it underfits (high bias). With no limit each point gets its own leaf: training error 0, and the worst error on fresh test labels at the same x . That is overfitting (high variance).

2 Train error falls; test error turns up



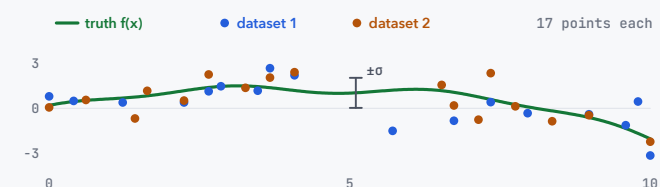
Averaged over 2000 datasets. Test MSE is lowest at depth 2 and climbs to 2.00, twice the noise, for the full tree. Held-out accuracy on the decision trees II sheet has the same shape, upside down. Pick the depth on validation data (the cross-validation sheet).

3 Bias is the offset, variance the spread



The slides' dartboard, drawn with real fits. Each dart is one model trained on its own random dataset, placed at its errors at $x = 4.08$ (across) and $x = 8.98$ (up). Centre: zero error; red ring: the average dart; one dart lands off each high-variance board. A constant misses the same way every time, and its darts lie on a diagonal: one number shifts both errors together.

4 One fixed truth, many possible datasets

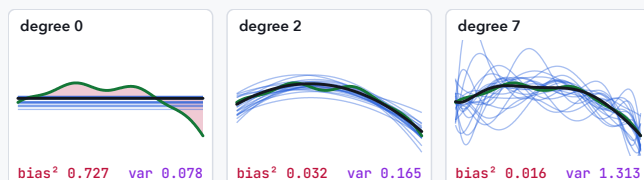


Labels are $y = f(x) + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$: zero mean, constant variance, independent across points and of the training set. The training set $\mathcal{D}$ is random, so the fit $\hat{f}$ is random too; $E[\hat{f}(x)]$ is its average over datasets.

$$\text{Bias}(x) = E[\hat{f}(x)] - f(x)$$

$$\text{Var}(x) = E[(\hat{f}(x) - E[\hat{f}(x)])^2]$$

5 Many fits show bias and variance



Each dataset: 17 of the 50 points (both ends and 15 at random), fresh noise, as in the notebook. Blue: fits to 20 datasets; black: the average of 2000; red band: bias. The constant has a wide band and tight fits; degree 2 nearly follows the truth; degree 7 bends to each dataset's noise, most at the ends.

6 Every model family has a complexity knob

	simpler: high bias	flexible: high variance
tree depth	1	no limit
polynomial degree	0	9
k in k-NN	large k	k = 1
ridge penalty λ	large	0
features used	few	many

The slides' profiles: small- k k-NN, deep trees and unregularised deep networks have low bias and high variance; linear regression and large- k k-NN the reverse. A random forest has lower variance than one tree.

Expected error = bias² + variance + noise

The lecture's derivation at one test input x , then a check with 2000 simulated datasets per model.

7 Step 1: add and subtract $f(x)$

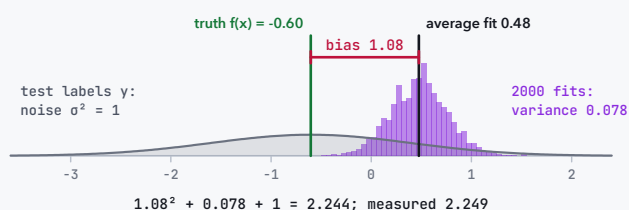
Fix x ; write $f = f(x)$ and $\hat{f} = \hat{f}(x)$. E averages over the training set $\mathcal{D}$ and the test label y .

$$E[(y - \hat{f})^2] = E\left[\left(\underbrace{y - f}_{\varepsilon} + \underbrace{f - \hat{f}}_{\text{model error}}\right)^2\right]$$

$$= E[\varepsilon^2] + 2E[\varepsilon(f - \hat{f})] + E[(f - \hat{f})^2]$$

$E[\varepsilon^2] = \text{Var}(\varepsilon) + (E[\varepsilon])^2 = \sigma^2$. The test noise is independent of $\mathcal{D}$, so the middle term is $2E[\varepsilon] \cdot E[f - \hat{f}] = 0$.

8 Step 2: add and subtract the average fit



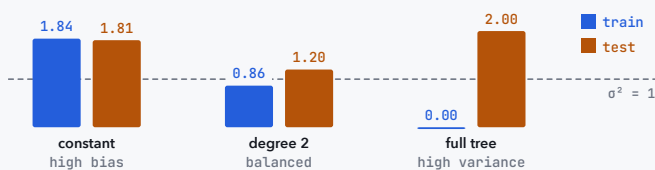
Let $\alpha = f - E[\hat{f}]$, a fixed number, and $\beta = E[\hat{f}] - \hat{f}$, whose mean is 0. Then

$$E[(f - \hat{f})^2] = \underbrace{E[\alpha^2]}_{\text{Bias}^2} + \underbrace{2\alpha E[\beta]}_0 + \underbrace{E[\beta^2]}_{\text{Var}(\hat{f})}$$

$$\text{so } E[(y - \hat{f}(x))^2] = \sigma^2 + \text{Bias}(x)^2 + \text{Var}(\hat{f}(x)).$$

Above: the constant model at $x = 8.98$. Purple: its predictions over 2000 datasets; grey: the test label.

9 Diagnose from training and test error



High bias: training error high, test error close to it. Try a more flexible model, more features, a deeper tree, less regularisation. **High variance:** training error far below test. Try more data, a shallower or pruned tree, regularisation, or averaging many models (the ensembles sheet).

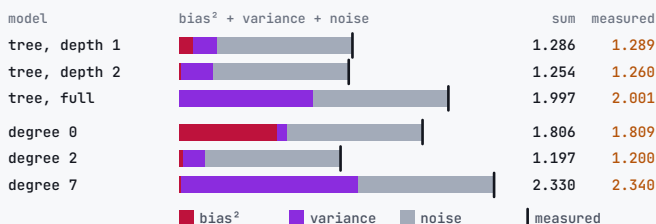
13 Pause and think

Can you measure bias on real data? Not directly: f is unknown and you have one dataset. Compare training and validation error.

Why does the full tree score 2.00? Each leaf holds one noisy label: bias 0, variance σ^2 . The test label adds another σ^2 .

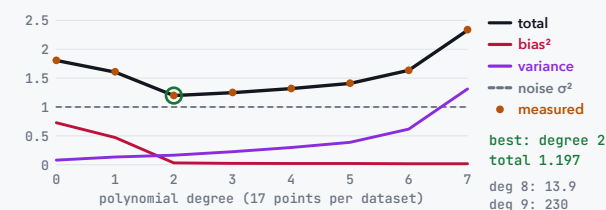
Choose depth by test error? No, that tunes on the test set. Use validation data or cross-validation.

10 The parts add up to the measured error



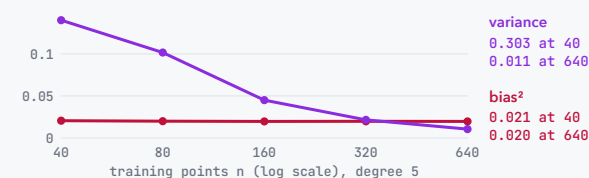
Bias² and variance from 2000 fits per model, noise $\sigma^2 = 1$, averaged over the 50 test inputs. Each sum is within 0.5% of the MSE on fresh test labels (black tick). Noise is the floor: no expected error goes below 1.

11 Total error is U-shaped in complexity



Bias² drops from 0.727 (constant) to 0.032 (degree 2), then barely moves. Variance grows from 0.078 to 1.313 at degree 7, then blows up with only 17 points. The best model, degree 2, keeps a little bias to avoid a lot of variance.

12 More data cuts variance, not bias



Degree 5 fitted to n random points. Doubling n about halves the variance; bias² stays at 0.020, since it belongs to the model family. More data lets you afford a more flexible model. The exception is a tree with no depth limit: one label per leaf keeps its variance at σ^2 (1.006 with 50 points, 0.993 with 400), so it still needs a depth limit or pruning.

error = bias² + variance + noise · flexibility trades bias for variance · tune complexity on validation data

Before: the decision trees II sheet; next: the cross-validation sheet. Source: lecture slides "Bias-variance and cross validation" and "The bias-variance tradeoff: a deep dive", and the bias-variance notebooks (nipunbatra.github.io/ml-teaching). Curve $f(x) = 0.2 \sin x + 0.2 \cos 2x + 0.6x - 0.05x^2 - 0.003x^3$ from the notebook; all runs are ours, seeded.